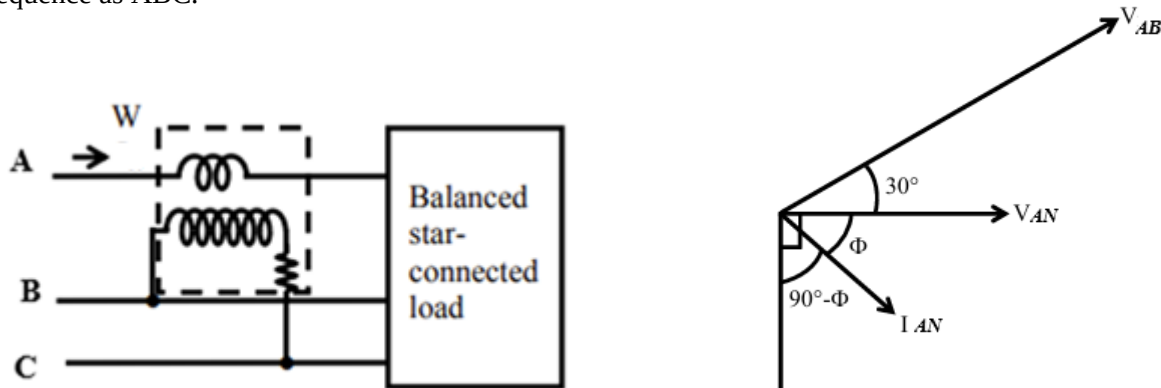


1. Calculate the readings of the wattmeter (W) connected as shown in Fig. The load is a balanced star-connected one, with impedance of $(2+j15) \Omega$ per phase, fed from a three-phase, 400 V, balanced supply, with the phase sequence as ABC. [5]



$$\text{Line voltage} = |V_L| = 400 \text{ V}$$

$$\text{Per phase impedance } Z_p = 20 + j15 = 25 \angle +36.87^\circ \Omega$$

$$\text{Magnitude of phase voltage: } |V_p| = |V_L|/\sqrt{3} = 400/\sqrt{3} = 231 \text{ V}$$

$$\text{Taking AN as reference, } V_{AN} = 231 \angle 0^\circ; \quad V_{BN} = 231 \angle -120^\circ; \quad V_{CN} = 231 \angle +120^\circ$$

$$I_{AN} = I_A = \frac{V_{AN}}{Z_p} = (231.0 \angle 0^\circ) / (25.0 \angle +36.87^\circ) = 9.24 \angle -36.87^\circ \text{ A}$$

$$V_{AB} = 400 \angle +30^\circ$$

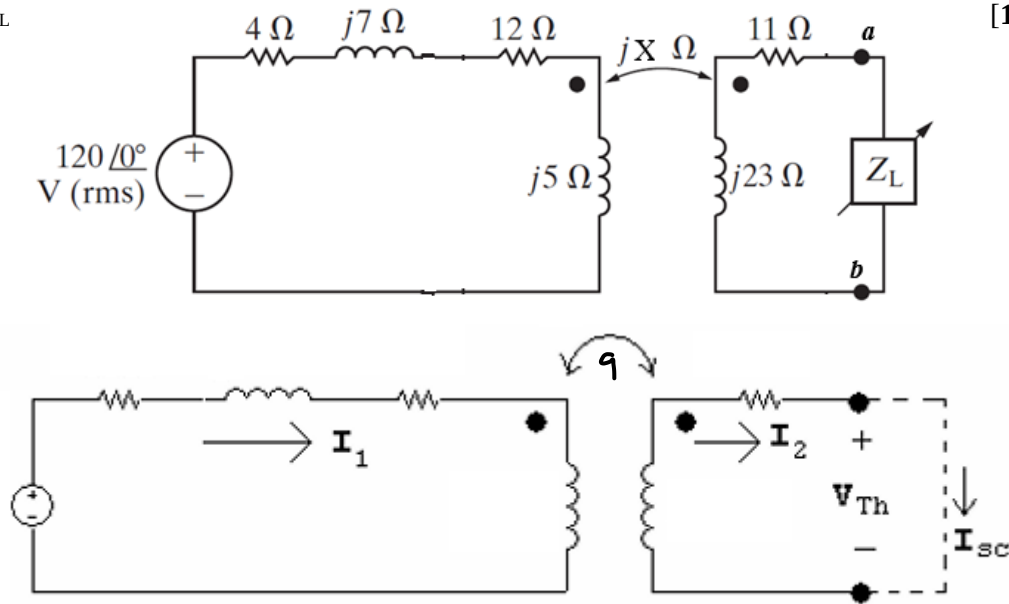
The voltage coil of the wattmeter is connected between the lines B and C and the current coil in line A

$$V_{BC} = 400 \angle -90^\circ \text{ V} \quad \text{Hence the wattmeter reading is } V_{BC} \times I_A \cos(\angle V_{BC}, \angle I_A)$$

```
x=9; % x=7,8,9,10
vth=(120/(16+12i))*x*i;% Vth (2 marks)
m=abs(vth);
phi=rad2deg(angle(vth));
a=[16+12i -x*i;-x*i 11+23i];
v=[120;0];
ic=inv(a)*v;
isc=ic(2,1);
zth=vth/isc;
zm=abs(zth);
phz=rad2deg(angle(zth));
z1=conj(zth); % ZL (4 marks)
i2=m/(zth+z1);
p=i2^2*real(z1);% Pmax max. power transferred to ZL (4 marks)
```

2.

Find the Thevenin's equivalent voltage across the terminals a, b if $X = 7, 8, 9, 10$. Also the maximum power transferred to Z_L [10]



Open the terminals a and b to find Thevenin's voltage $v_{th} = (120 / (16 + 12i)) * x * i$; where $x = 7, 8, 9, 10$

To find I_{sc} , short circuit a, b

$$V_{TH} = jX I_1$$

Mesh equations are $[a][ic] = v$

Where ic is the short circuit current. $Z_{th} = V_{th} / I_{sc}$

Find the current through Z_{th} using the Thevenin's circuit. $P_{max} = i^2 R$

```
x=9; % x=7,8,9,10
vth=(120/(16+12i))*x*i;% Vth (2 marks)
m=abs(vth);
phi=rad2deg(angle(vth));
a=[16+12i -x*i;-x*i 11+23i];
v=[120;0];
ic=inv(a)*v;
isc=ic(2,1);
zth=vth/isc;
zm=abs(zth);
phz=rad2deg(angle(zth));
z1=conj(zth); % ZL (4 marks)
i2=m/(zth+z1);
p=i2^2*real(z1);% Pmax max. power transferred to ZL (4 marks)
```